CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 6 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. What is the primary objective of organizing a 'Health Run' in a community? (A) To raise funds for school infrastructure (B) To raise awareness about health, physical fitness, and an active lifestyle (C) To identify elite athletes for national teams (D) To distribute prizes and scholarships *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q2. Which specific postural deformity is characterized by the collapsing or dropping of the medial longitudinal arch of the foot? (A) Flat Foot (B) Knock Knees (C) Bow Legs (D) Club Foot *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q3. In the context of Yoga, which of the following asanas is famously translated as the "Tree Pose" and helps in improving body balance? (A) Tadasana (B) Halasana (C) Vajrasana (D) Shalabhasana *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [Practice] | Tag: High***

Q4. Which specialized professional works specifically to help Children with Special Needs (CWSN) improve their ability to perform Activities of Daily Living (ADLs) independently? (A) Sports Psychologist (B) Speech Therapist (C) Occupational Therapist (D) Dietician *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q5. Which of the following is a Water-Soluble vitamin? (A) Vitamin A (B) Vitamin D (C) Vitamin C (D) Vitamin K *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q6. If 12 teams are participating in an inter-school competition conducted purely on a Single League basis, how many total matches will be played? (A) 11 matches (B) 66 matches (C) 132 matches (D) 144 matches *Unit 1: Management of Sporting Events | [Practice] | Tag: High***

Q7. A menstrual dysfunction characterized by extremely painful menstrual cramps and severe pelvic pain in female athletes is medically termed as: (A) Menarche (B) Amenorrhea (C) Dysmenorrhea (D) Osteoporosis *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q8. Which essential trace mineral is directly responsible for the production of hemoglobin in red blood cells, ensuring oxygen reaches working muscles? (A) Calcium (B) Iron (C) Zinc (D) Potassium *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q9. In the SAI Khelo India Fitness Test battery, what does the Flamingo Balance Test primarily measure? (A) Dynamic Balance (B) Static Balance (C) Lower Body Strength (D) Reaction Time *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q10. The number of times the human heart beats per minute at complete rest is known as the: (A) Stroke Volume (B) Cardiac Output (C) Resting Heart Rate (D) Vital Capacity *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q11. The term 'Inertia' in sports biomechanics represents an object's tendency to: (A) Accelerate instantly (B) Change its direction of motion (C) Resist any change in its state of rest or uniform motion (D) Generate friction *Unit 8: Biomechanics and Sports | [PYQ 2024] | Tag: Very High***

Q12. An injury where a bone breaks completely but does not puncture or pierce through the skin is called a: (A) Compound Fracture (B) Simple (Closed) Fracture (C) Comminuted Fracture (D) Open Fracture *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q13. In a knock-out fixture, the privilege given to a top-seeded team to skip the preliminary rounds and directly play in the Quarter-Finals or Semi-Finals is called: (A) Bye (B) Regular Seeding (C) Special Seeding (D) Tie-Breaker *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q14. The Rikli & Jones 'Arm Curl Test' for senior citizens is administered to measure: (A) Upper body flexibility (B) Upper body strength and endurance (C) Lower body strength (D) Aerobic endurance *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q15. Which of the following forces actively acts to slow down a rolling soccer ball on a grass field? (A) Air resistance only (B) Rolling friction (C) Static friction (D) Centripetal force *Unit 8: Biomechanics and Sports | [PYQ] | Tag: High***

Q16. Assertion (A): Skipping meals is an effective, scientifically recommended strategy for fast and healthy weight loss in athletes. **Reason (R):** Skipping meals drastically lowers the body's Basal Metabolic Rate (BMR) and causes severe nutritional deficiencies and fatigue. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition | [Practice] | Tag: Very High***

Q17. Assertion (A): In a knock-out tournament, losing a single match permanently eliminates the team from the competition. **Reason (R):** Knock-out tournaments save time and organizers' money compared to lengthy round-robin league tournaments. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events | [Practice] | Tag: High***

Q18. Assertion (A): A sumo wrestler usually spreads his legs wide apart and bends his knees while standing in the ring. **Reason (R):** Widening the base of support and lowering the center of gravity significantly increases an athlete's physical stability and equilibrium. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | [PYQ 2024] | Tag: Very High***

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. Differentiate briefly between a 'Bye' and a 'Special Seeding' in a tournament fixture. *Unit 1: Management of Sporting Events | [PYQ] | Tag: High***

Q20. Suggest any two yoga asanas that are highly beneficial in preventing and curing 'Asthma'. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q21. State any two practical strategies a physical education teacher can adopt to make sports and physical activities accessible for Children With Special Needs (CWSN). *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: High***

Q22. A student's calculated BMI is 26.5. What does this value specifically indicate regarding their weight category? What advice would you give them? *Unit 5: Sports & Nutrition | [Practice] | Tag: Very High***

Q23. Define 'Contusion'. How does it visibly manifest on an athlete's body? *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q24. What is meant by the 'Trajectory' of a projectile? State any one factor that alters this path. *Unit 8: Biomechanics and Sports | [PYQ] | Tag: High***

SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

Q25. Outline the major pre-tournament responsibilities of the Publicity Committee and the Boarding & Lodging Committee. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q26. Enlist the physiological causes that lead to the postural deformity known as 'Round Shoulders'. Suggest any two corrective exercises. *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q27. Explain the step-by-step procedure and mention any two health benefits of practicing *Pawanmuktasana*. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: High***

Q28. Discuss the crucial functions of 'Proteins' and 'Fats' as essential macronutrients in an athlete's diet. *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q29. Differentiate clearly between a 'Simple (Closed) Fracture' and a 'Compound (Open) Fracture'. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: High***

Q30. State Newton's Second Law of Motion (Law of Acceleration). Give one practical example of its application in sports. *Unit 8: Biomechanics and Sports | [PYQ 2024] | Tag: Very High***

SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

Q31. The local sports club of a city is organizing a district-level Table Tennis tournament on a knock-out basis. A total of 11 teams have officially registered. The organizing committee sits down to mathematically calculate the fixture structure to ensure the tournament runs smoothly without any mismatches. **(i)** How many total matches will be played to declare the champion? **(ii)** Calculate the number of teams that will be placed in the upper half of the fixture. **(iii)** How many 'Byes' will be awarded in the entire fixture? **(iv)** In which specific round will the teams that received a bye play their first match? **OR** Write the formula used to calculate the number of rounds when N is not a power of 2. *Unit 1: Management of Sporting Events | [Practice] | Tag: Very High***

Q32. Rahul is preparing for a state-level marathon (42 km). Three days before the race, his coach strictly alters his diet, advising him to eat heavy portions of whole wheat pasta, brown rice, and potatoes while drastically cutting down on fats and fibrous foods. **(i)** Which specific macronutrient is the coach asking Rahul to consume in large quantities? **(ii)** What is the primary purpose of loading this specific nutrient before a marathon? **(iii)** Which micro-mineral should Rahul ensure he consumes to maintain high hemoglobin levels for optimal oxygen transport during the long run? **(iv)** Name the diet-planning technique where endurance athletes intentionally maximize their glycogen stores before an event. **OR** Out of Carbohydrates, Proteins, and Fats, which one yields the highest energy per gram (9 kcal/g)? *Unit 5: Sports & Nutrition | [Practice] | Tag: High***

Q33. During the Olympics high jump event, an athlete uses the famous "Fosbury Flop" technique. As he approaches the bar, he aggressively pushes his take-off foot into the ground. In the air, he arches his back backward over the bar, allowing his body to clear the height while his center of gravity actually passes *under* the bar. **(i)** Pushing the take-off foot aggressively into the ground to propel the body upward is a classic demonstration of which of Newton's Laws? **(ii)** What is the 'Center of Gravity'? **(iii)** In the air, the athlete is entirely under the influence of which invisible downward force? **(iv)** When the high jumper lands safely on the thick foam mat, what type of anatomical movement occurs when he bends his knees to absorb the shock? **OR** State Newton's First Law of Motion. *Unit 8: Biomechanics and Sports | [PYQ] | Tag: Very High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 4)

Word limit: 200-300 words.

Q34. What is a Fixture? Draw a neat and comprehensive Knock-out fixture for 17 teams. Clearly show all your mathematical calculations including the number of matches, total byes, rounds, and the distribution of byes in the upper and lower halves. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q35. Children with Special Needs (CWSN) require a multi-disciplinary approach to participate safely in physical activities. Discuss in detail the specific roles of any four professionals (e.g., Counselor, Physiotherapist, Occupational Therapist, Speech Therapist, or Special Educator) in assisting CWSN. *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q36. The Rikli & Jones Senior Citizen Fitness Test evaluates the functional fitness of elderly individuals. Detail the administration procedure and the specific fitness component measured by the following three test items: (a) Chair Stand Test, (b) Back Scratch Test, and (c) 8-Foot Up and Go Test. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q37. Define 'Cardiorespiratory Endurance'. Regular, intensive training brings profound adaptations to the human body. Elaborate on any four long-term physiological effects of regular exercise on the Cardiovascular System. *Unit 7: Physiology & Injuries in Sport | [PYQ 2024] | Tag: High***

Answer Key

Q1. (B) To raise awareness about health, physical fitness, and an active lifestyle **Q2.** (A) Flat Foot **Q3.** (A) Tadasana **Q4.** (C) Occupational Therapist **Q5.** (C) Vitamin C **Q6.** (B) 66 matches (Formula: $N(N-1)/2 = 12 \times 11/2 = 66$) **Q7.** (C) Dysmenorrhea **Q8.** (B) Iron **Q9.** (B) Static Balance **Q10.** (C) Resting Heart Rate **Q11.** (C) Resist any change in its state of rest or uniform motion **Q12.** (B) Simple (Closed) Fracture **Q13.** (C) Special Seeding **Q14.** (B) Upper body strength and endurance **Q15.** (B) Rolling friction **Q16.** (D) (A) is false, but (R) is true. **Q17.** (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (Reason does not explain *why* a team is eliminated, it only explains an advantage of the format). **Q18.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

Q19. Bye: A privilege given to a team (usually by drawing lots) exempting them from playing in the first round. **Special Seeding:** A higher privilege where top-seeded teams are directly placed in the Quarter-Finals or Semi-Finals, skipping all preliminary rounds. **Q20.** 1. *Gomukhasana* (Cow Face Pose) expands the chest and lungs. 2. *Chakrasana* (Wheel Pose) stretches the respiratory organs and clears airways. **Q21.** 1. **Modify Equipment:** Use auditory balls (with bells) for visually impaired children or lighter bats/racquets. 2. **Modify Rules:** Allow extra time, extra trials, or reduce the size of the playing court to accommodate their physical limits. **Q22.** A BMI of 26.5 falls into the **Overweight** category (range $25 - 29.9$). **Advice:** The student should adopt a balanced diet (caloric deficit) and engage in regular aerobic physical activities (like swimming or running) to reduce body fat. **Q23. Contusion** is a soft tissue injury (bruise) usually caused by a direct blunt hit (like a sports equipment striking the skin). It manifests as swelling and a blue/black discoloration on the skin due to internal bleeding (ruptured blood capillaries) without breaking the outer epidermis. **Q24. Trajectory** is the parabolic flight path followed by a projectile (like a thrown javelin or shot-put) in the air. **Factor:** Angle of release, initial velocity, or air resistance.

Q25. Publicity Committee: Responsible for advertising the sports event well in advance through newspapers, social media, and banners to attract teams and spectators. **Boarding & Lodging Committee:** Responsible for arranging safe accommodation (stay) and hygienic meals for outstation teams, athletes, and officials throughout the tournament. **Q26. Causes:** 1. Weak chest and back muscles. 2. Habitual poor posture while sitting, reading, or carrying heavy backpacks on one shoulder. **Corrective Exercises:** 1. Performing *Chakrasana* or *Dhanurasana* to stretch the chest muscles outward. 2. Holding a horizontal bar and hanging regularly. **Q27. Procedure:** Lie flat on the back (supine). Inhale, bend both knees, and bring them close to the chest. Interlock the fingers around the shins to hug the knees. Exhale, lift the head, and try to touch the nose/forehead to the knees. Hold, then release. **Benefits:** 1. Relieves constipation and releases trapped abdominal gas. 2. Reduces lower back fat and strengthens core muscles. **Q28. Proteins:** Known as the "building blocks" (4 kcal/g); essential for repairing micro-tears in muscle tissues post-workout, aiding in muscle recovery and hypertrophy (growth). **Fats:** A highly concentrated energy source (9 kcal/g); cushions vital internal organs, regulates body temperature, and is necessary for the absorption of fat-soluble vitamins (A, D, E, K). **Q29.** A **Simple (Closed) Fracture** is an internal break in the bone where the broken ends do not pierce or break through the outer skin layer. A **Compound (Open) Fracture** is a severe injury where the broken bone violently splinters and pierces through the skin, exposing the bone to the external environment, carrying a high risk of severe infection. **Q30. Newton's Second Law (Law of Acceleration):** States that the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass ($F = ma$). **Example:** In a shot-put event, an athlete must apply maximum explosive muscular force to the heavy iron ball to ensure it accelerates rapidly and covers a greater horizontal distance.

Q31. (i) Total Matches = $N - 1 = 11 - 1 = 10$ matches. (ii) Upper half teams = $(N + 1)/2 = 12/2 = 6$ teams. (iii) Total Byes = Next power of 2 (16) - $11 = 5$ byes. (iv) Second Round. **OR** Next highest power of 2 is 2^n. The number of rounds is equal to n (Here, $16 = 2^4$, so 4 rounds). **Q32.** (i) Carbohydrates. (ii) To maximize glycogen stores in the muscles and liver, providing a massive reservoir of sustained energy for the long endurance race. (iii) Iron. (iv) Carbohydrate Loading (Carbo-loading). **OR** Fats. **Q33.** (i) Newton's Third Law of Motion (Law of Action and Reaction). (ii) Center of Gravity is the imaginary central point around which the total weight of the athlete's body is evenly balanced. (iii) Gravity. (iv) Flexion (angle between the thigh and calf decreases). **OR** An object at rest or in uniform motion remains so unless acted upon by an external force.

Q34. A **Fixture** is a systematically drawn schedule of matches outlining which team plays against whom, at what time, and on which court. **Calculations for 17 Teams:** * Matches: $N - 1 = 17 - 1 = 16$. * Total Byes: Next power of 2 (32) - $17 = 15$ byes. * Teams in Upper Half: $(N + 1)/2 = 18/2 = 9$. * Teams in Lower Half: $(N - 1)/2 = 16/2 = 8$. * Byes in Upper Half: $(Nb - 1)/2 = 14/2 = 7$ byes. * Byes in Lower Half: $(Nb + 1)/2 = 16/2 = 8$ byes. * Rounds: $32 = 2^5$, so 5 Rounds. *(Evaluator Note: Bracket drawn with 17 teams. 9 in upper, 8 in lower. 7 byes placed in upper half, 8 in lower half following bottom-top-bottom-top rule. Teams without byes play Round 1. Winners advance up to Round 5 Finals).* **Q35.** 1. **Physiotherapist:** Uses physical exercises, massages, and mobility therapies to improve the CWSN's gross motor skills, joint flexibility, and muscle strength. 2. **Occupational Therapist:** Focuses on improving fine motor skills (like gripping a pen or a ball) helping the child perform Activities of Daily Living (ADLs) independently. 3. **Speech Therapist:** Evaluates and treats communication disorders, helping CWSN overcome stutters or speech impediments so they can interact with teammates. 4. **Special Educator:** Modifies the standard physical education curriculum and adapts teaching methods to suit the unique cognitive and physical learning pace of the child. **Q36. (a) Chair Stand Test:** *Measures:* Lower body strength. *Procedure:* Participant sits on a chair with arms crossed. On 'go', they stand up fully and sit back down. Score is the max complete stands in 30 seconds. **(b) Back Scratch Test:** *Measures:* Upper body flexibility (shoulder). *Procedure:* One hand reaches over the shoulder behind the back, the other reaches up from the lower back. Participant tries to touch fingers. Score is the distance (or overlap) between middle fingers. **(c) 8-Foot Up and Go Test:** *Measures:* Agility and dynamic balance. *Procedure:* Participant sits in a chair, stands up on 'go', walks 8 feet around a cone, and sits back down. Score is the time taken in seconds. **Q37. Cardiorespiratory Endurance** is the ability of the heart, lungs, and blood vessels to supply oxygen-rich blood to the working muscles efficiently during prolonged physical activity. **Long-term Effects on Cardiovascular System:** 1. **Cardiac Hypertrophy:** The heart muscle (especially the left ventricle) increases in size and thickness, becoming a much stronger pump. 2. **Decreased Resting Heart Rate (Bradycardia):** Because the heart is stronger, it pumps more blood per beat, requiring fewer beats per minute while at rest. 3. **Increased Stroke Volume and Cardiac Output:** The volume of blood ejected per beat (stroke volume) and per minute (cardiac output) significantly increases during heavy exercise, delivering more oxygen to muscles. 4. **Faster Recovery Rate:** Post-exercise, an athlete's heart rate returns to its normal resting baseline much quicker than an untrained individual.
